Name: Uzair Ali
Reference Code: ARZ-2026-EBPP

Q1)Write a Governance Memo comparing:

- **Vulnerability Assessment**
- **Penetration Testing**
- **Red Teaming**

Feature	**Vulnerability Assessment**	**Penetration Testing**	**Red Team**
Goal	Find vulnerabilities	Exploit vulnerabilities	Simulate attackers
Exploitation	No	Yes	Yes
Stealth	No	Limited	High
Duration	Hours	Days	Weeks
Automation	High	Medium	Low
Business Impact	Estimated	Demonstrated	Fully simulated

Q2)Design a segmented lab architecture including:

- **Tester Machine**
- **Target Machine**
- **Monitoring System**
- **IP Address Plan**
- **Snapshot Policy**
- **Network Restrictions**

Ans)

Tester Machine

Kali Linux
Purpose:

- Nmap
- Burp Suite
- Nikto
- Gobuster

- Hydra

IP:
192.168.19.128

Target Machine

Windows 10
Purpose:

- Test environment
- Web browser
- Future deployment of OWASP Juice Shop or DVWA

IP:
192.168.19.129

Monitoring System

Kali Linux
Purpose:

- Monitor network traffic
- Capture packets
- Verify communication between tester and target

IP:
192.168.19.129

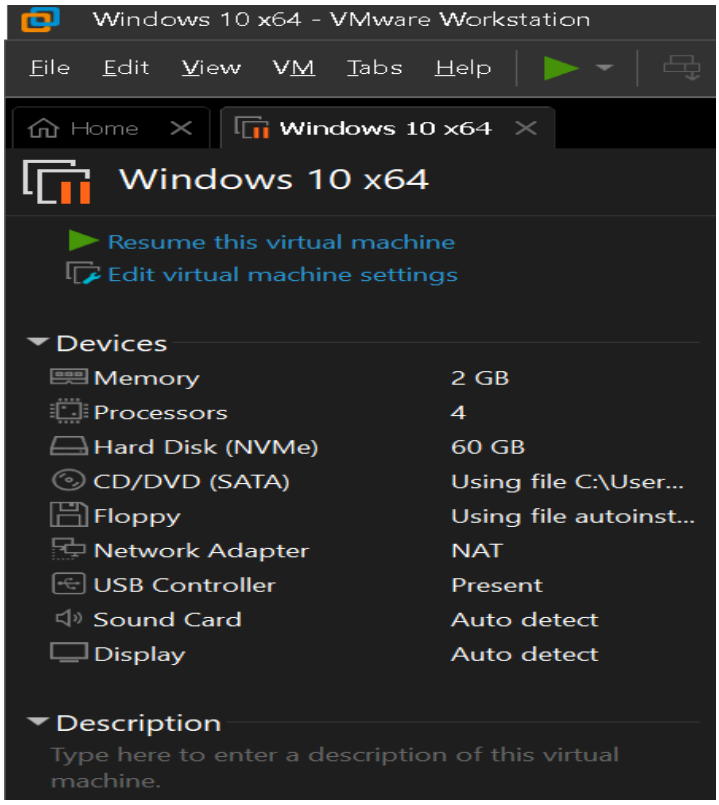
Q3)Create a Testing Methodology Matrix mapping:

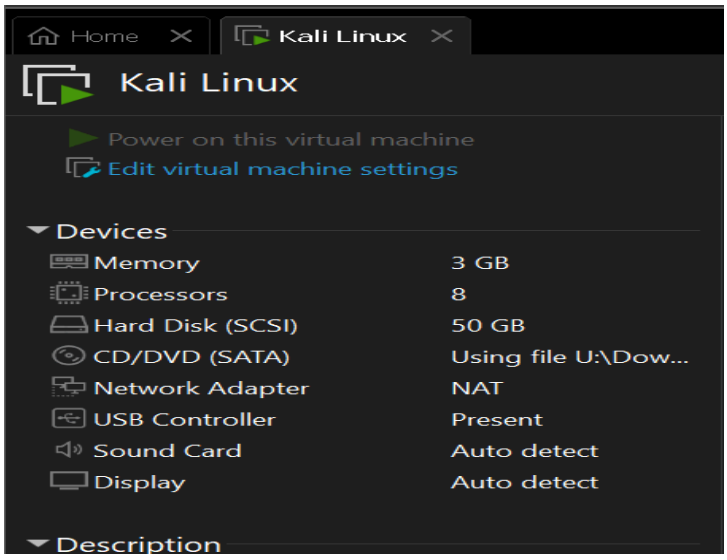
- **NIST SP 800-115**
- **OWASP WSTG**
- **MITRE ATT&CK to your lab activities.**

Framework	Lab Activity
NIST SP 800-115	Planning, Reconnaissance, Scanning, Exploitation, Reporting
OWASP WSTG	Information Gathering, Authentication Testing, Input Validation, Session Testing
MITRE ATT&CK	Reconnaissance, Initial Access, Discovery, Credential Access

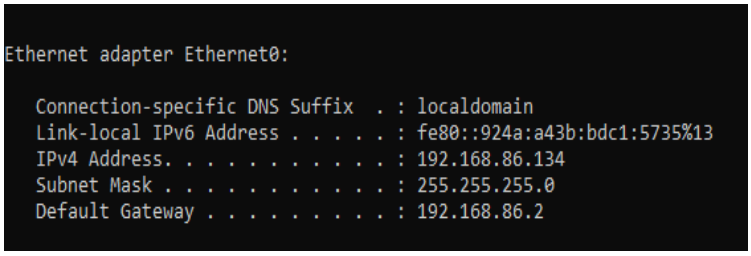
Q4)Build the lab and provide evidence of:

Ans)VM Configuration :





Network Isolation :



```
-$ ip a
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
 qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
 default qlen 1000
    link/ether 00:0c:29:f7:89:c8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.86.129/24 brd 192.168.86.255 scope global dynamic noprefixroute
eth0
       valid_lft 1626sec preferred_lft 1626sec
    inet6 fe80::20c:29ff:fef7:89c8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

4)Proof that only lab systems can communicate

```
┌──(redhunter㉿kali)-[~]
└─$ ping 192.168.19.129
PING 192.168.19.129 (192.168.19.129) 56(84) bytes of data.
64 bytes from 192.168.19.129: icmp_seq=1 ttl=128 time=8.02 ms
64 bytes from 192.168.19.129: icmp_seq=2 ttl=128 time=3.41 ms
64 bytes from 192.168.19.129: icmp_seq=3 ttl=128 time=5.09 ms
64 bytes from 192.168.19.129: icmp_seq=4 ttl=128 time=2.02 ms
64 bytes from 192.168.19.129: icmp_seq=5 ttl=128 time=1.16 ms
64 bytes from 192.168.19.129: icmp_seq=6 ttl=128 time=1.52 ms
64 bytes from 192.168.19.129: icmp_seq=7 ttl=128 time=0.943 ms
64 bytes from 192.168.19.129: icmp_seq=8 ttl=128 time=2.71 ms
64 bytes from 192.168.19.129: icmp_seq=9 ttl=128 time=1.04 ms
64 bytes from 192.168.19.129: icmp_seq=10 ttl=128 time=1.52 ms
64 bytes from 192.168.19.129: icmp_seq=11 ttl=128 time=2.76 ms
64 bytes from 192.168.19.129: icmp_seq=12 ttl=128 time=1.11 ms
64 bytes from 192.168.19.129: icmp_seq=13 ttl=128 time=1.82 ms
64 bytes from 192.168.19.129: icmp_seq=14 ttl=128 time=1.46 ms
```

```
:\Users\Window 10>ping 192.168.19.128

inging 192.168.19.128 with 32 bytes of data:
eply from 192.168.19.128: bytes=32 time=1ms TTL=64
eply from 192.168.19.128: bytes=32 time<1ms TTL=64
eply from 192.168.19.128: bytes=32 time=1ms TTL=64
eply from 192.168.19.128: bytes=32 time=1ms TTL=64

ing statistics for 192.168.19.128:
   Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
   Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Q5) Penetration Testing Kickoff Pack

Ans)

Scope Sheet

In Scope

- Kali Linux Tester Machine
- Windows 10 Target Machine
- Host-Only Network (VMnet1)
- Network enumeration and security testing
- Authorized security tools (Nmap, Burp Suite, Nikto, Gobuster, Hydra)

Out of Scope

- Host Operating System
- Public Internet
- Production Systems
- Third-party devices or services
- Any system outside the lab network

Rules of Engagement (ROE)

The following rules must be followed during penetration testing:

1. Testing will only be performed on authorized lab machines.
2. No testing will be conducted outside the defined scope.
3. Denial-of-Service (DoS) attacks are prohibited.
4. Destructive payloads or malware are not allowed.
5. All vulnerabilities will be documented responsibly.
6. Any critical issue will be reported immediately.
7. Confidentiality of collected information must be maintained

Risk Register

Risk	Impact	Likelihood	Mitigation
VM Crash	Medium	Medium	Restore VM Snapshot
Incorrect Network Configuration	Medium	Medium	Verify network settings before testing
Data Loss	High	Low	Backup important files
Unauthorized Testing	High	Low	Stay within approved scope
Tool Misconfiguration	Medium	Medium	Verify tool configuration before execution

Communication Plan

Event	Responsible Person	Communication Method
Testing Start	Tester	Notify Instructor
Critical Vulnerability Found	Tester	Immediate Notification

Daily Progress	Tester	Daily Status Update
Testing Complete	Tester	Submit Final Report (PDF)

Stop Conditions

Testing must stop immediately if:

- An unauthorized system is discovered.
- A production environment is accidentally reached.
- A virtual machine becomes unstable or crashes.
- The instructor requests testing to stop.
- A critical system failure occurs.
- Testing exceeds the approved scope.